

RRC-Raptor-V4H

- Display module

V0.0.1

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i. Revision List

Description	Page	Version	Date	Author
New creation	-	0.0.1	Sep, 18 2024	Felix Hsu

ii. HASHTAG

#V4H

#kernel

#Display

#libdrm

#eDP

#HDMI

#Fakra

#SerDes

1. Overview

1.1 Objectives

This manual is intended to introduce the display module (this module) of RRC V4H.

1.2 Scope

This manual is only used for developing and validating the RRC V4H display module.

1.3 Functionality

Function list supported to this module is as follows.

Function	Support status
Output connector	HDMI x1 Fakra connector x4 mini DP x1

1.3.1 Display Resolution

Supported resolution of this module is as follow.

NOTICE: Using any resolution that hasn't been evaluated might produce results different from what is expected and is not recommended.

Table 1-1: Supported resolution (RRC SOM + CB)

Display resolution	eDP	HDMI	Fakra
VGA (640x480)	V		
SVGA (800x600)	V		
XGA (1024x768)	V	V	
WXGA (1280x720)	V	V	
WXGA (1280x768)	V	V	V
FHD (1920x1080)	V	V	V

WUXGA (1920x1200)	V	V	
QHD (2560x1440)	V		

1.3.2 Connector

Supported connector of this module is as follows. For more details, please refer to document “104_RRC-Raptor-V4H_Connector”

Table 1-2: Connector Number (RRC CB)

Output signal	Connector's number
eDP	CN8
HDMI	CN9
GMSL2	CN10, CN11, CN37, CN38

1.3.3 DIP switch

Two DIP switches can be used to control the direction of the DSI signal. For more details, please refer to document “103_RRC-Raptor-V4H_Switch”

Table 1-3: DIP switch information (RRC CB)

Switch index	Content
SW2	Control DSI-0 signal to SerDes (CN10/CN11) / eDP (CN8)
SW44	Control DSI-1 signal to SerDes (CN37/CN38) / HDMI (CN9)

1.4 Relative Document

The following Table 1-4 indicates the reference documents of RRC V4H display module.

Table 1-4: Table Heading (RRC SOM + CB)

Manufacture	Title	Edition
Retronix Inc. Tech	103_RRC-Raptor-V4H_Switch	Rev.0.0.2
Retronix Inc. Tech	104_RRC-Raptor-V4H_Connector	Rev.0.0.2

2. Terminology

The following Table 2-1 indicates the terminology related to this manual.

Table 2-1: Terminology

Terms	Description
CB	Carrier Board
DU	Display Unit on R-car series
DRM	Direct Rendering Manager

3 . Operating Environment

3.1 Hardware Environment

The following Table 3-1 indicates the hardware needed for RRC V4H display module.

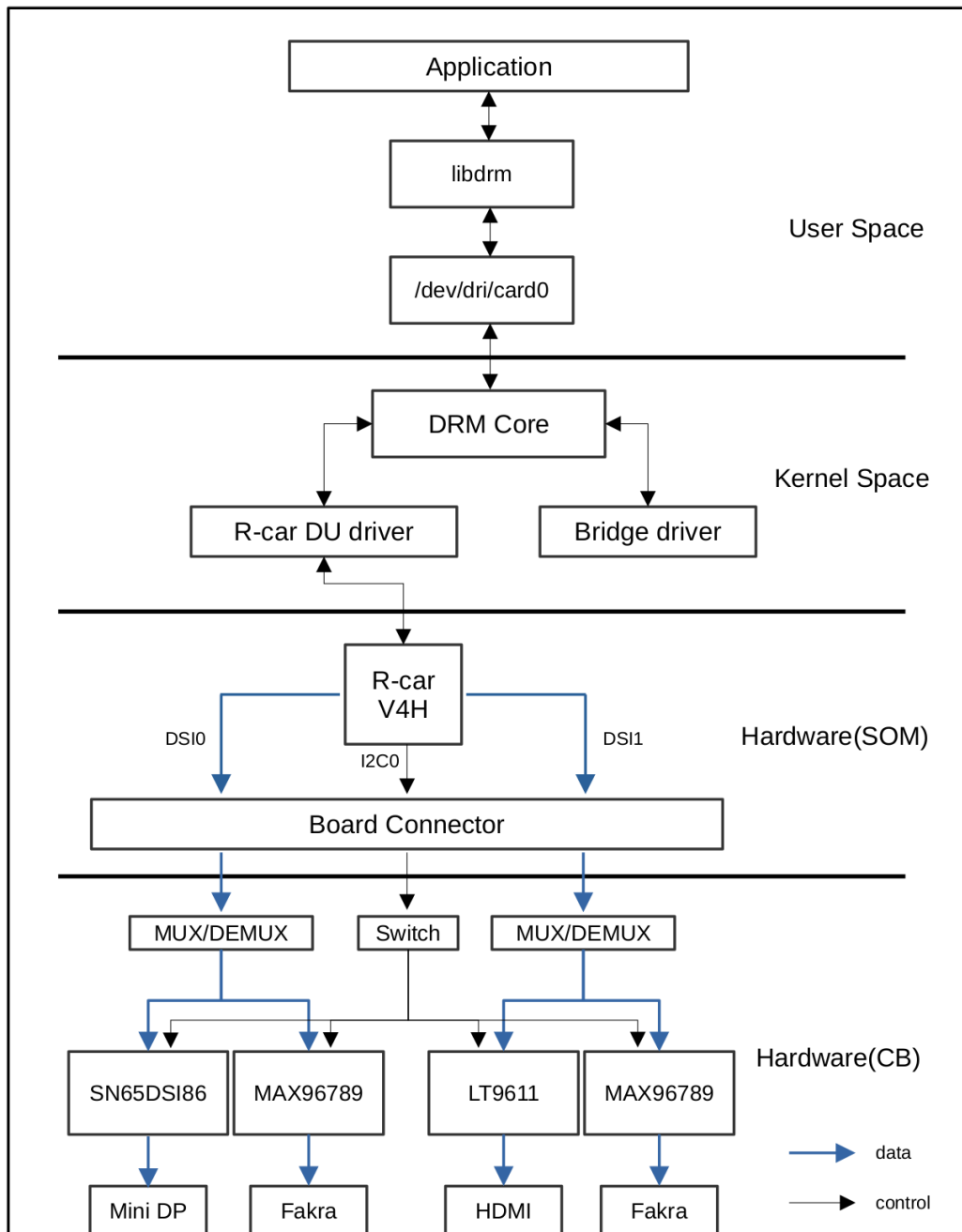
Table 3-1: Hardware specification

Name	Manufacture
RRC V4H SOM	Retronix Tech Inc.
RRC V4H CB	Retronix Tech Inc.

3.2 Module configuration

The following Figure 3-1 indicates the configuration of RRC V4H display module.

Figure 3-1: Module configuration



4 . Software Interface

The software interface of RRC is based on Linux.

4.1 Userspace

Device node of this module is shown below.

Table 4-1: Device node major and minor number

Name	Major	Minor
/dev/dri/card0	226	0

4.1.1 libdrm

This module corresponds to the libdrm v2.4.104.

For details of libdrm, please refers to the following.

- Download site
<https://gitlab.freedesktop.org/mesa/drm>

‘/tests/modetest/modetest.c’ attached to libdrm is a sample test program and can serve as a reference of using libdrm. To instruction on how to use libdrm, please refers to source code on the download site. To use the sample test, please refers to the example below.

Example A: Check options modetest support

```
# modetest --help
```

Example B: Dump all info of DRM component

```
# modetest -M rcar-du
```

Example C: Set a mode with specific crtc and connector

```
# modetest -M rcar-du -s <conn-id>@<crtc-id>:1920x1080
```

Format:

```
modetest -M <module name> -s <conn-id>@<crtc-id>:<mode>[-<vrefresh>][@<format>]
```

Description:

Option	Description
<module name>	This module use rcar-du. Must be given.
<conn-id>	specific connector id. Use Example B to dump all the connector id.
<crtc-id>	specific crtc id. Use Example B to dump all the crtc id.
<mode>	<x resolution>x<y resolution> e.g. 1920x1080
<vrefresh>	Option. Refers to Framerate.
<format>	Option. Refers to Bits per pixel.

4.1.2 Unsupported Function

The libdrm APIs in which this display module is not supported are described.

Please do not use the API in the following.

Table 4-2: List of unsupported function for this module

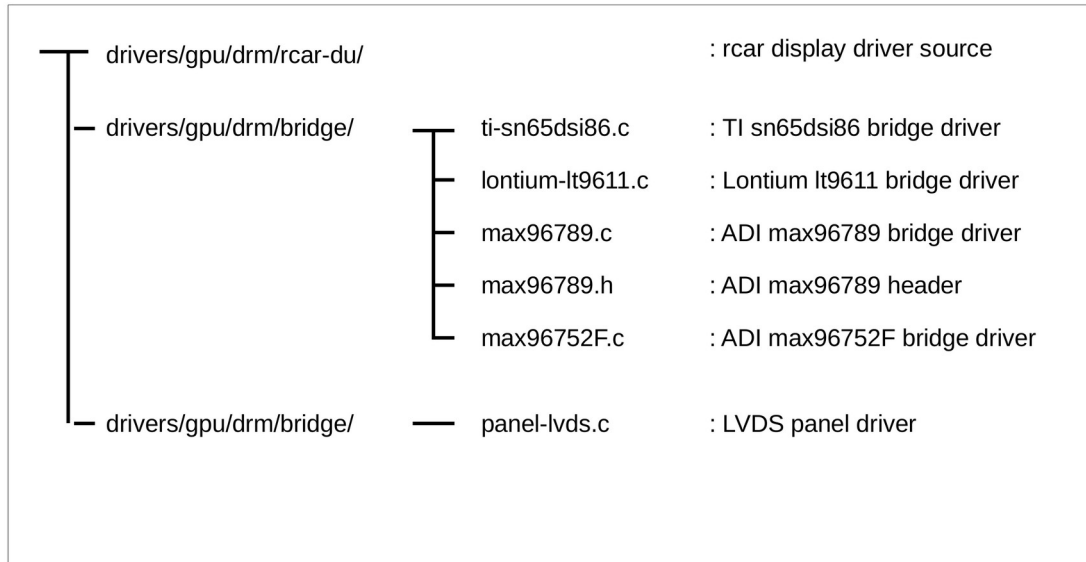
Function name
drmModeCrtcGetGamma, drmModeCrtcSetGamma
drmModeSetCursor, drmModeSetCursor2
drmCreateDrawable, drmDestroyDrawable
drmUpdateDrawableInfo
drmAgp
drmFinish
drmGetInterruptFromBusID

[Notice]

- Scaling can't be performed with the Display driver when using drmModeSetPlane()

4.2 Kernel Space

4.2.1 Directory



4.2.2 Device Tree – HDMI and eDP

Since the display module supply eDP, HDMI, and GMSL2 connection, it is necessary to predefine the display link in the device tree in order to load the correct configuration and driver. The device tree is define in the following path:

Path <kernel-source>/arch/arm64/boot/dts/renesas/r8a779g0-raptor.dts

Please follow the step to enable appropriate link for demand. **By default, eDP and HDMI are enabled.**

[To use the eDP (CN8) connection]

```
&dsi0 {
.....
        endpoint {
+           remote-endpoint = <&sn65dsi86_in0>;
+           data-lanes = <1 2 3 4>;
        };
};
```

[To use the HDMI (CN9) connection]

```
&dsi1 {
.....
        endpoint {
+           remote-endpoint = <&lt9611_in>;
        };
};
```

4.2.3 Device Tree – GMSL

By default the Serializer on RRC is disabled. To use it, the setting below shall be followed.

[Change DSI-0 endpoint to point to MAX96789 (CN10/11)]

```
&dsi0 {
.....
        endpoint {
+           remote-endpoint = <&u28_max96789_in>;
-           remote-endpoint = <&sn65dsi86_in0>;
        data-lanes = <1 2 3 4>;
        };
};
```

```
U28: max96789@60 {
-     status = "disabled";
+     status = "okay"
};
```

[Change DSI-1 endpoint to point to MAX96789 (CN37/38)]

```
&dsi1 {
.....
        endpoint {
+           remote-endpoint = <&u181_max96789_in>;
-           remote-endpoint = <&lt9611_in>;
        };
};
```

```
U181: max96789@40 {
-     status = "disabled";
+     status = "okay"
};
```

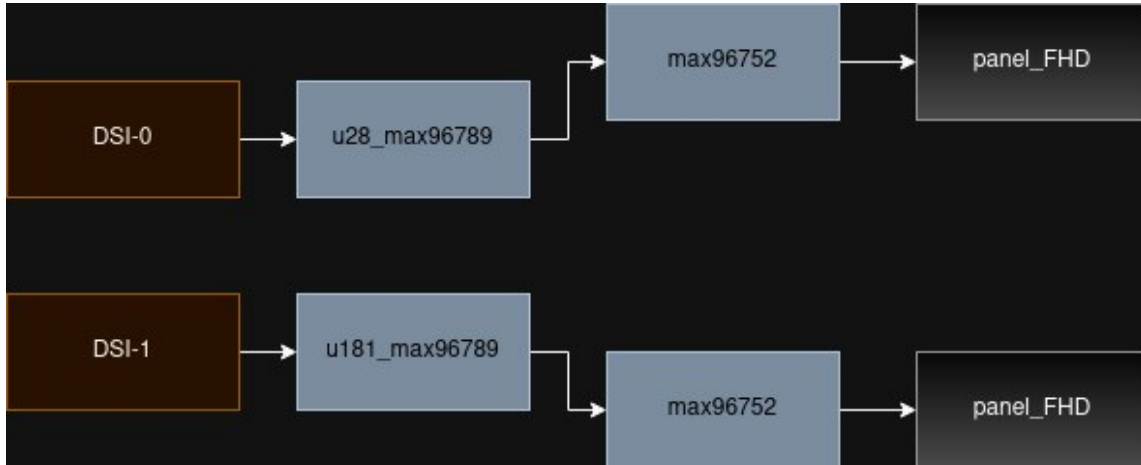
The port number of MAX96789 in the device tree has a specific meaning. Table 4-3 outlines the corresponding relationship with the hardware:

Table 4-3: Port number corresponding with the hardware

Chip number	Port number	Hardware
U28	0	Endpoint between MAX96789 (0x60) and DSI-0 transceiver
	1	Endpoint between MAX96789 (0x60) and sink device (CN10)
	2	Endpoint between MAX96789 (0x60) and sink device (CN11)
U181	0	Endpoint between MAX96789 (0x40) and DSI-1 transceiver
	1	Endpoint between MAX96789 (0x40) and sink device (CN37)
	2	Endpoint between MAX96789 (0x40) and sink device (CN38)

Since the SerDes doesn't have EDID detect functionality, there are two ways to correctly set the display timing: device tree pre-defined, or driver default.

4.2.3.1 Device Tree Pre-defined



Description

The display pipeline is like figure above. Each MAX96789 is followed by a deserializer connects to a panel. The most important point is that every block in the figure has its own driver for configuration and can't be omitted. The structure of the device tree will be as follows:

```

1  / {
2  .....
3  panel {
4      compatible = "panel-lvds";
5      .....
6      panel-timing {
7          clock-frequency = <>;
8          hactive = <>;
9          vactive = <>;
10         hsync-len = <>;
11         hfront-porch = <>;
12         hback-porch = <>;
13         vfront-porch = <>;
14         vback-porch = <>;
15         vsync-len = <>;
16     };
17
18     ports {
19         .....
20         panel_in: endpoint {
21             remote-endpoint = <&max96752F_out>;
22         };
23     };
24 };
25
26
27 &i2c_bus {
28     U28: max96789@60 {
29         compatible = "maxim,max96789";
30         .....
31         ports {
32             .....
33             u28_max96789_out0: endpoint {
34                 remote-endpoint = <&max96752F_in>;
35             };
36         };
37     };
38
39     max96752F@48 {
40         compatible = "maxim,max96752F";
41         .....
42         ports {
43             .....
44             max96752F_in: endpoint {
45                 remote-endpoint = <&u28_max96789_out0>;
46             };
47             max96752F_out: endpoint {
48                 remote-endpoint = <&panel_in>;
49             };
50         };
51     };
52 };

```

Parse timing

Decide your panel timing here

Configure the chip

Configure the chip

Note that if the deserializer in your device doesn't need any configurations by driver, it would be fine to connect the endpoint "u28_max96789_out" to "panel_in" directly. For more details, please check "r8a779g0-raptor.dts" for example.

4.2.3.2 Driver Default



Description

The display pipeline is like the figure above. The MAX96789 is followed by an external converter, which means it has already handled all the deserializer configuration in the device. The display sink is a monitor that supports common and wide range of resolution. In this case, the device tree can be as follows:

```

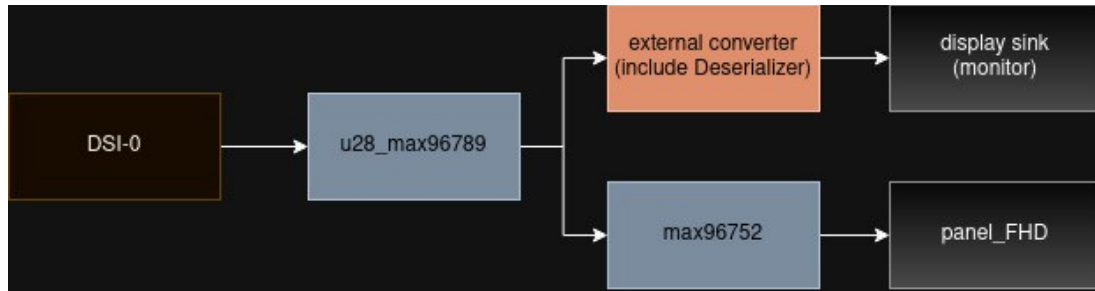
1 > &i2c_bus {
2 >     U28: max96789@60 {
3 >         compatible = "maxim,max96789";
4 >         .....
5 >         ports {
6 >             .....
7 >         };
8 >     };
9 > };
  
```

Specifying any endpoint and connection is not needed, and the default resolution setting is decided by max96789's driver shown below. For other application that needs different resolution setting, please refer to "4.2.4 Command line parameters"

```

static int max96789_bridge_get_modes(struct drm_bridge *bridge,
                                     struct drm_connector *connector)
{
    int count;
    count = drm_add_modes_noedid(connector, 8192, 8192);
    drm_set_preferred_mode(connector, 1920, 1080);
    return count;
}
  
```

4.2.3.3 Mixed-case and Limitation



Description

There may be cases in which applications connect an external converter to one port and a panel to the other. The following points must be considered carefully in this scenario:

- According to the DRM architecture, each encoder (in this case, the DSI encoder) is followed by only one connector. This means that although MAX96789 can split the DSI signal into two ports, both ports (CN10/CN11 for DSI-0 or CN37/CN38 for DSI-1) are treated as a single connector in the kernel.
- Since both hardware ports are treated as a single connector in the kernel, the output signal timing will be the same from both ports (CN10/CN11 for DSI-0 or CN37/CN38 for DSI-1).
- In MAX96789's driver, the priority of a pre-defined timing is higher than that of the driver's default. Therefore, in this scenario, the kernel only accepts timing defined in the device tree. An image can be displayed on the monitor and panel only if the monitor supports the same timing as the panel.

4.2.4 Command line parameters

Additional parameters which are environment variables of u-boot could be set to parse the specific pixel format information into kernel when start-up. By default, the eDP and HDMI drivers use the EDID read technique, which means it's not necessary to set command line parameters. Additionally, the SerDes uses fixed timing predefined in the Device Tree, so specifying video information in the command line is generally unnecessary. Therefore, this chapter is optional.

Please add the parameter to bootargs with the following format:

Format video=[name-of-connector]:[h-resolution x v-resolution][R][-<bpp>][@<refresh-rate>][i]

Table 4-3: Description of command line parameter “video”

Connector name	HDMI-A-1(HDMI0 output), DP-1(DP output)
[h-resolution x v-resolution]	Depends on resolution supported by monitor
bpp	Bits per pixel. e.g. 24 (RGB888)
Refresh-rate	Please specify refresh rate
R option	If 'R' is specified, do a 'reduced blanking' calculation for digital displays.
i option	If 'i' is specified, calculate for an interlaced mode.

Example

Please add the underline part to the bootargs.

[To use eDP]

```
[FHD]: RGB888, 60fps  
bootargs=rw root=/dev/nfs nfsroot=192.168.0.1:/export/rfs ip=192.168.0.20 video=DP-1:1920x1080-24@60
```

[To use HDMI]

```
[FHD]: RGB888, 60fps  
video=HDMI-A-1:1920x1080-24@60
```

[To use Serdes with pre-defined timing]

It's not necessary to use command line when the timing is defined in the device tree. For detailed format information, please refer to the **panel_wxga** or **panel_1080p** nodes in the r8a779g0-raptor.dts file

[To use Serdes with driver default timing]

```
[FHD]: RGB888, 60fps  
video=DSI-1:1920x1080-24@60
```

If both SerDes use driver default timing, then the other connector name will be “DSI-2”

[To use both eDP and HDMI]

```
[FHD]: RGB888, 60fps  
video=DP-1:1920x1080-24@60 video=HDMI-A-1:1920x1080-24@60
```